Zewail City of Science, Technology and Innovation
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CSAI 203 -Fall 2026

Introduction to Software Engineering

<Academic Course & Schedule Planner>

<SRS Document>

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1.Introduction

1.1 Purpose.

The Purpose of the **Software Requirements Specification** (SRS) is to understand the requirements, scope and objective of the Academic Course & Schedule Planner web application. The requirements contain Functional, Nonfunctional and External Interface Requirements.|

By using this document, The readers will know the description of the project, understanding the features of the project, and will know how to use the product.

This System will aim to offer academic planning by helping the student register for his courses, make the timetable, and check prerequisite, calculated GPA, and section details. It helps the instructor to check the section details and give a grade. It helps the registration to modify sections and courses.

1.2 Scope.

The Academic Course & Schedule Planner is a web-based application designed to facilitate academic schedule planning for students and management of academic course information for administrators.

The system will be a web app where students build semester conflict-free timetables, track prerequisite requirements, and calculate GPAs, while admins manage course offerings, capacities, and prerequisites.

The system will be used to manage academic schedules and planning. The application will allow students to easily make the right choices regarding their studies while providing administrators with the tools necessary to edit and verify the information.

The key aspects of the system will include the management of course schedules, ensuring that there are no schedule conflicts, course prerequisites are met, and the number of enrolled students does not exceed limits. Additionally, the system will calculate the GPA for students based on the selected courses and credit points. The system can be interconnected to external university websites and programs (e.g., student information system, official website with course selection) if needed.

1.3 Definitions, Acronyms, and Abbreviations

The following table defines the technical terms, acronyms, and abbreviations used throughout this document.

Term	**Definition**
GPA	Grade Point Average: A numerical representation of a student's academic performance.

UI	User Interface; the visual elements through which users interact with the system.
CGPA	Cumulative Grade Point Average: The final grade of all graded semesters
VS	Visual Studio codes

1.4 References.

Dr. Mohamed Rakha course Pdfs.

IEEE 830-1998 standard

1.5 Overview.

The document is organized as following order:

- Section 2 (Overall description): it will contain the discerption of the product and its functions, the users of the product, the stakeholders.
- Section 3 (Specific Requirements): It will contain functional and non-functional requirements, and use case model, domain model, product interface.
- Section 4 (Appendices): It will contain a data dictionary and the glossary.

2. Overall Description.

2.1 Product Perspective.

The Academic Course & Schedule Planner will be a standalone web application that will help in managing universities student and admission and instructors.

2.2 Product Functions.

- Account registration
- Searching for courses
- GPA Calculation
- Administration to courses, students, instructors.
- Timetable Construction.

2.3 User Classes and Characteristics

- **Student**

The students' experience level may be beginner to intermediate to deal with their responsibilities such as checking prerequisite, planning schedule and making their timetables.

- **Administrators**

The Administrators experience level may be intermediate to deal with their responsibilities such as grading the students and resolving schedule conflicts.

2.4 Software Stakeholders

The Stakeholders of this product will be student, instructor, academic advisor, and administrator.

2.5 Operating Environment

This product will be operated on Chrome 90

2.6 Design and Implementation Constraints.

The development of Academic Course & Schedule Planner will be subject to the following constraints.

- **Programming language:** The required programing language will be Python, and it is not allowed to use another one.
- **Programming Tools:** The implementation process will use VS code.
- **Data Storage:** The system will use dictionaries and lists to manage data with JSON files to save data.

2.7 User Documentation.

The system will provide the following documentation to help users understand and use the system without any problems.

- **User Manual**
  A simple guide explains how to interact with the system successfully.
- **Help Section**
  Section called help in system help users to deal with system without issues.

2.8 Assumptions and Dependencies.

- Having internet access.
- Having a valid university account.
- The student has basic computer skills.
- Course information is updated and provided by the administrator.
- The system depends on OS to run the application

3. Specific Requirements.

3.1 Functional Requirements.

- FR1: Registration.
  The system shall allow user registration with email verification.

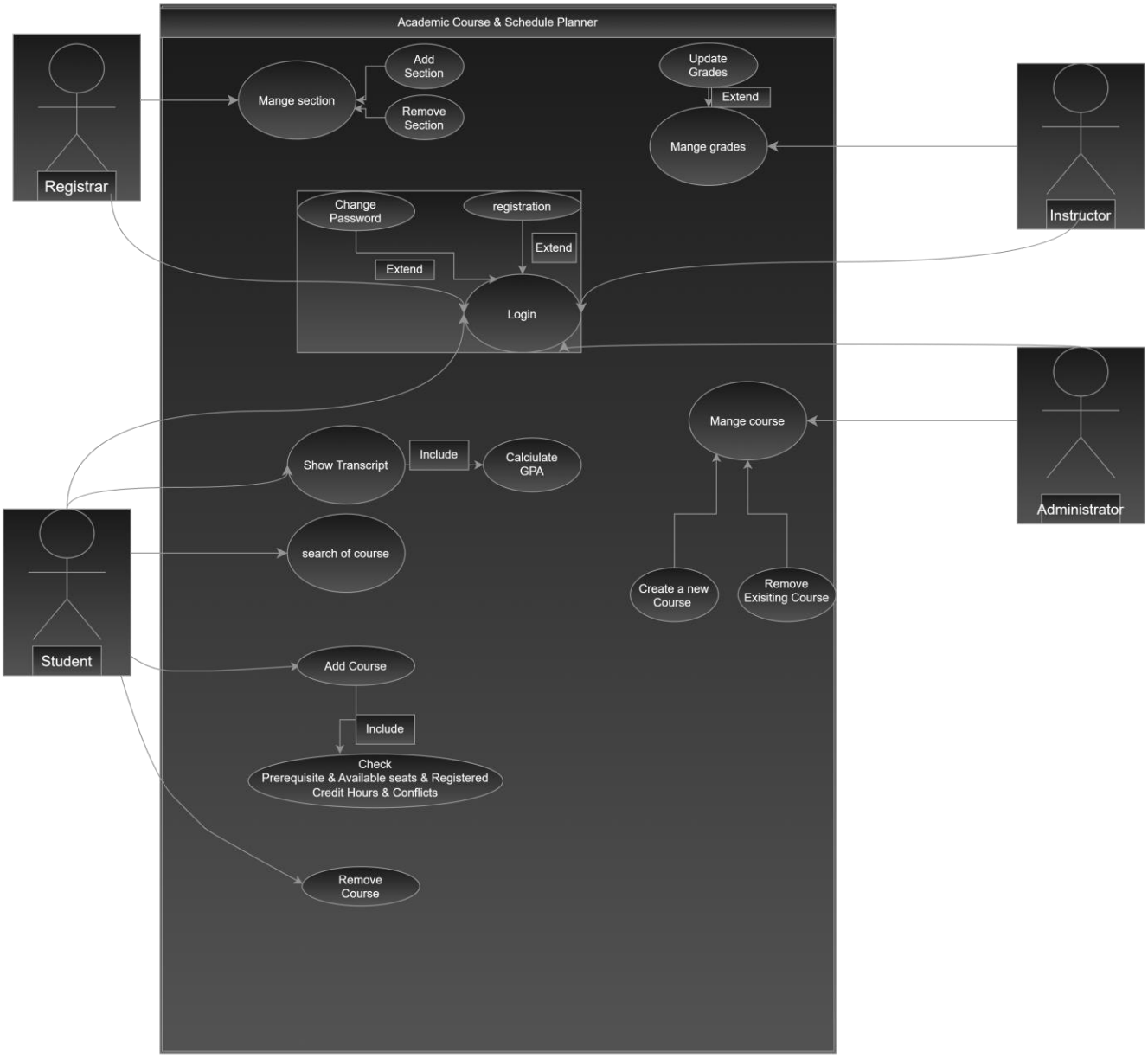
- FR2: Login.
  The System shall allow user to access the app with a verified account and a correct password.
- FR3: Calculating_GPA.
  The system shall calculate the total GPA of the Student grades after each semester and after each year.
- FR4: Add_A_New_Section.
  The System shall allow the registration to add a course section so that the student can register in that section
- FR5 :Remove_Existing_section.
  The System shall allow the registration to remove any section they added.
- FR6: Create_A_New_Course.
  The System shall allow the admission to create a course for the students while adding its details.
- FR7: Delete_Existing_Course.
  The System shall allow the admission to delete any existing course.
- FR8: Add_Course_Section.
  The System shall allow the student to add course section to their timetable.
- FR9: Drop_Course_Section.
  The System shall allow the student to drop a course section.
- FR10: Show_Timetable.
  The System shall allow the student to see all of his courses in a Timetable.
- FR11: Give_Grade.
  The instructor gives the grades for all students enrolled in this course
- FR12: Update_Grade.
  The instructor can change the grade for a student if there was a mistake
- FR13: Credit_Hour_Limit_Using_GPA.
  The System will calculate the student maximum credit hour limit using his GPA.
- FR14: Check_Prerequisite.
  The system should make sure that the student passed the prerequisite courses before the registration of a specific course.
- FR 15: Check_Section_Conflict.
  The system should check if there is conflict before registration and advising the student if he should go to the academic advisor to solve thiis conflict or not.
- FR 16: Check_If_The_Seat_Is_Available.
  The system should check if there are any seats in a section before registration.
- FR 17: Show_Transcript
  The student can see all of his passed courses with his grade
- FR 18: Profile_management.
  The student can manage his profile like changing the password.
- FR 19: Check_Credithours.

The system checks the number of credit hours the student has registered and if the number of registered credit_hours is bigger than assigned credit limit.

- FR 20: Search_For_Course_Section.
  The system helps the student search through the courses.

3.2 Use Case Model

3.2.1 Use Case Diagram



3.2.2 Use Case Description

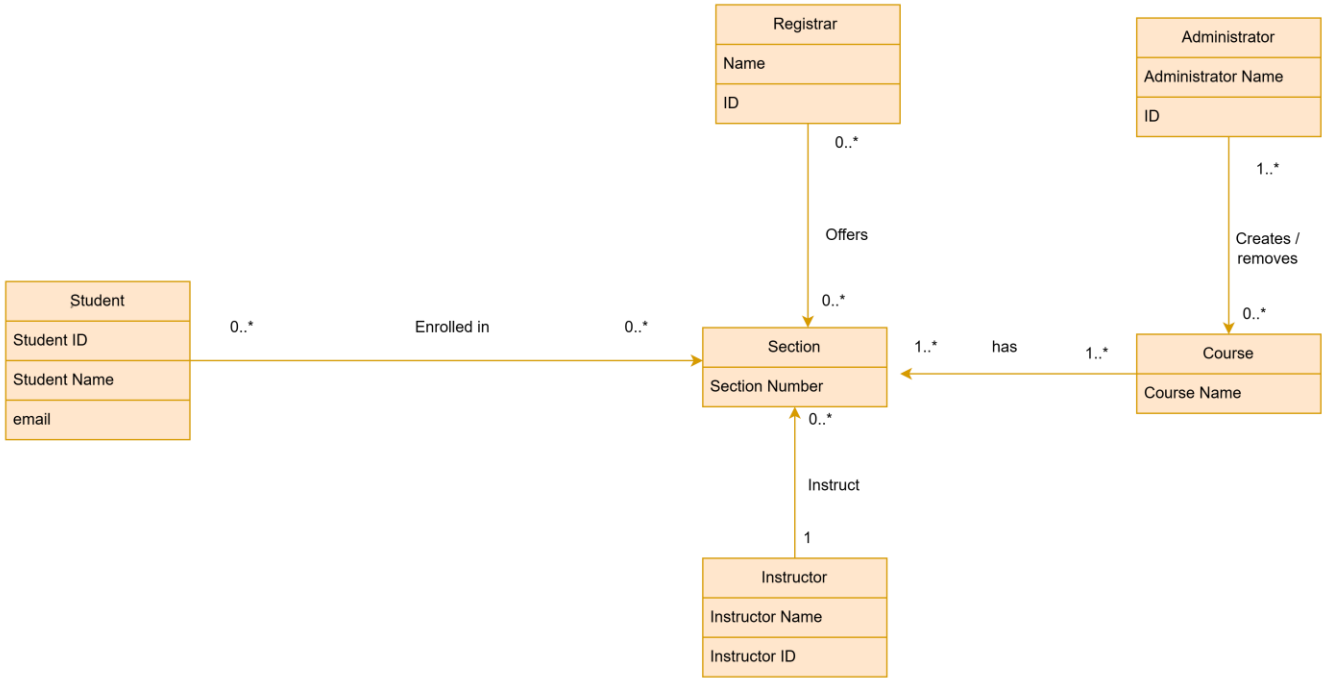
UC ID	UC name	Description	Actors	Preconditions	Postconditions	Flow
UC_1	Login	All users log in to their account by email and password to the website	Student, Administrator, Instructor, Registrar	A valid Email and Password	Log in completed	The user Will be asked to enter their email and password. If they are true, the user can access the website successfully. The user will be asked to enter the email and password again.
UC_2	Manage section	The Registrar will Add or remove courses sections in website so students can register these.	Registrar	Registrar makes sure to add or remove this section	The course section has been added or removed	The registrar adds or removes the courses sections according to academic planner
UC_3	Manage grades	The instructor adds the grades of each student and can update them too.	Instructor	Marks for every student of 100	Student has his grade	At the end of semester each student has their mark of 100 and then with Calculate GPA to get the final

						grade.
UC_4	Show Transcript	The student can show their transcript with name, ID, and grades of all courses.	Student	The grades of all registered courses	The student can see their grades for all semesters and CGPA	Student click on “Show transcript” And then the student can see their CGPA and grades of all courses
UC_5	Manage Course	The administrator creates or removes a course to the university curriculum	administrator	Make sure of adding or deleting this course	Course added or deleted	Administrator adds a new course with all information needed and can delete existing one also.
UC_6	Register course	Student wants to register a course after the registrar offers this course	student	Make sure that the student has passed all the prerequisites of this course, there are available seats still offered in this section, the student did not pass the specific credit hours per term, and there are no conflict	The course has been registered successfully	Student wants to register courses. If he passed the prerequisite courses, has no conflicts, has enough credit hours, there is one seat at least, the student can register for this course. if not, the student will get an error

				hours.		message
UC_7	Drop Course	The student wants to drop a course after he has registered for it.	student	Make sure that the student has registered for it already.	Course dropped successfully	Student wants to drop a course after registration it so he can drop the course successfully
UC_8	Search A course	Student wants to search about course to register it or know a little information about it	student	Make sure that course is existing to show a result	Wanted course showed successfully	Students want to search about course so then enter the name and will get what they want

3.3 Domain Model

3.3.1 Conceptual Class Diagram



3.3.2 Class Describtion

Class Name	Description	Key Attributes	Associations
Student	One of the users for this system and the one that this system build for	Student Name, ID, email	Enrolls in Section
Administrator	One that administrate the academic process	Administrator Name, ID	The one that can create or remove a course from academic planner
Course	Represent the courses that administrator adds	Course Name	The administrator creates the course in academic planner then students can register the course using the sections.

Section	Represents the sections of courses registrar offers	Section Number	The registrar offers the sections of different courses so students can register in these sections.
Instructor	Represents Dr of each section	Instructor Name, ID	He instructs the different sections of all courses and puts the grades
Registrar	Represents the person who offer sections	Registrar Name, ID	Registrar offers sections and chooses number of seats for each section

3.4 Nonfunctional Requirements

1. **Useability**
   Usability refers to How easy and comfortable the system is to use and understand.
   We can test it by asking students if they can register for a course by themselves without any assistance.

2. **Security**
   Security ensures that users' information is protected, and that unauthorized users cannot access restricted features such as students should only access their own academic information.
   We can test it by try accessing in wrong password

3. **Reliability**
   Reliability refers to the system`s ability to work correctly without unexpected failures.
   We can test it by restarting the system and verifying that saved data is still correct.

4. **Performance**
   Performance describes how fast the system responds to user actions and how efficiently it handles data such as searching for a course should be fast, Registration for a course should not take a long time.
   We can test it by measuring the time required to search or register a course

5. **Maintainability**
   Maintainability describes how easily developers can fix and update the system without affecting other parts.

We can test it by trying to add new features and checking whether existing features still work or not.

3.5 External Interface Requirements

3.5.1 User Interface.

The Academic Course & Schedule Planner will provide a simple user interface.
Login Page: allow the users to access their accounts successfully.
Main Page (student): This Page will have a navigation bar at the top of the page include my profile, transcript, register course and timetable.
each one of them has different page with different features to deal with it easily.

Main Page (instructor): This page will have a navigation bar at the top of the page include my courses, Course_Gardes.

Main Page (admission): This Page Will have navigation to creating and removing courses.

3.5.4 Communication Interface.

The most important protocols will be used is **https to make sure of our site safety.**

4 Appendices.

4.1 Appendix A: Data Dictionary.

Table Name	Description
student	Stores student information such as ID, Name, Email, Password
Instructor	Stores instructor information
Course	Stores course information
Registrar	Stores registrar information
Prerequisite	Stores the prerequisite courses required before registration

4.2 Appendix B: Glossary.

SRS: Software Requirements Specification
UI: User Interface
CGPA: Cumulative Grade Point Average